

Thesis presented to the Instituto Tecnológico de Aeronáutica, in partial fulfillment of the requirements for the degree of Doctor of Science in the Graduate Program of Engenharia Aeronáutica e Mecânica, Field of Sistemas Aeroespaciais e Mecatrônica.

**Daniel Vieira**

**ROBUST CONTROL OF LINEAR SYSTEMS WITH  
SWITCHED ACTUATORS SUBJECTED TO  
DWEELL-TIME CONSTRAINTS**

Thesis approved in its final version by signatories below:

Prof. Dr. Roberto Kawakami Harrop Galvão  
Advisor

Prof<sup>a</sup>. Dr<sup>a</sup>. ZZZZZZZZZZZZZZZZZZZZZ  
Co-advisor

Prof. Dr. ZZZZZZZZZZZZZZZZZZZZZ  
Dean for Graduate Education and Research

Campo Montenegro  
São José dos Campos, SP - Brazil  
2020

**Cataloging-in Publication Data**  
**Documentation and Information Division**

Vieira, Daniel  
Robust control of linear systems with Switched Actuators Subjected to Dwell-Time Constraints  
/ Daniel Vieira.  
São José dos Campos, 2020.  
39f.

Thesis of Doctor of Science – Course of Engenharia Aeronáutica e Mecânica. Area of Sistemas Aeroespaciais e Mecatrônica – Instituto Tecnológico de Aeronáutica, 2020. Advisor: Prof. Dr. Roberto Kawakami Harrop Galvão. Co-advisor: Prof<sup>a</sup>. Dr<sup>a</sup>. ZZZZZZZZZZZZZZZZZZZ.

1. Cupim. 2. Dilema. 3. Construção. I. Instituto Tecnológico de Aeronáutica. II. Title.

**BIBLIOGRAPHIC REFERENCE**

VIEIRA, Daniel. **Robust control of linear systems with Switched Actuators Subjected to Dwell-Time Constraints**. 2020. 39f. Thesis of Doctor of Science – Instituto Tecnológico de Aeronáutica, São José dos Campos.

**CESSION OF RIGHTS**

AUTHOR'S NAME: Daniel Vieira

PUBLICATION TITLE: Robust control of linear systems with Switched Actuators Subjected to Dwell-Time Constraints.

PUBLICATION KIND/YEAR: Thesis / 2020

It is granted to Instituto Tecnológico de Aeronáutica permission to reproduce copies of this thesis and to only loan or to sell copies for academic and scientific purposes. The author reserves other publication rights and no part of this thesis can be reproduced without the authorization of the author.

---

Daniel Vieira  
Pca Romao Gomes, 98  
12.233-066 – São José dos Campos–SP

# ROBUST CONTROL OF LINEAR SYSTEMS WITH SWITCHED ACTUATORS SUBJECTED TO DWELL-TIME CONSTRAINTS

**Daniel Vieira**

Thesis Committee Composition:

|                                       |                                |            |   |        |
|---------------------------------------|--------------------------------|------------|---|--------|
| Prof. Dr.                             | Alan Turing                    | Presidente | - | ITA    |
| Prof. Dr.                             | Roberto Kawakami Harrop Galvão | Advisor    | - | ITA    |
| Prof <sup>a</sup> . Dr <sup>a</sup> . | ZZZZZZZZZZZZZZZZZZZZ           | Co-advisor | - | OVNI   |
| Prof. Dr.                             | Linus Torwald                  |            | - | UXXX   |
| Prof. Dr.                             | Richard Stallman               |            | - | UYYY   |
| Prof. Dr.                             | Donald Duck                    |            | - | DYSNEY |
| Prof. Dr.                             | Mickey Mouse                   |            | - | DISNEY |

**ITA**

Aos amigos da Graduação e Pós-Graduação do ITA por motivarem tanto a criação deste template pelo Fábio Fagundes Silveira quanto por motivarem a mim e outras pessoas a atualizarem e aprimorarem este excelente trabalho.

# Acknowledgments

Primeiramente, gostaria de agradecer ao Dr. Donald E. Knuth, por ter desenvolvido o T<sub>E</sub>X.

Ao Dr. Leslie Lamport, por ter criado o L<sup>A</sup>T<sub>E</sub>X, facilitando muito a utilização do T<sub>E</sub>X, e assim, eu não ter que usar o Word.

Ao Prof. Dr. Meu Orientador, pela orientação e confiança depositada na realização deste trabalho.

Ao Dr. Nelson D'Ávila, por emprestar seu nome a essa importante via de trânsito na cidade de São José dos Campos.

Ah, já estava esquecendo... agradeço também, mais uma vez ao T<sub>E</sub>X, por ele não possuir vírus de macro :-)

*"I am and always will be the optimist.  
The hoper of far-flung hopes, and the dreamer of improbable dreams."*  
— THE ELEVENTH DOCTOR

# Resumo

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o sistema à trajetória cíclica alvo.

# Abstract

Switched actuators are difficult to handle because of some characteristic constraints of their nature. but they are efficient in numerous engineering applications, being applied, for example, in attitude control systems for space vehicles. Thus, knowing how to use this type of actuator is important and necessary. The main problem to be solved in this thesis is the design of a controller that guarantees the robustness of a system with switched actuators subjected to time constraints that limit the response window of their commands. In this approach, a limit cycle target trajectory search is performed and then the error of the states compared to the target trajectory is calculated. Then a predictive control technique is applied to correct the error and get the system to the cyclic target trajectory



# List of Figures

|                                                 |    |
|-------------------------------------------------|----|
| FIGURE 3.1 – Instantes de chaveamento . . . . . | 24 |
|-------------------------------------------------|----|

# List of Tables

# List of Abbreviations and Acronyms

|        |                                     |
|--------|-------------------------------------|
| CTq    | computed torque                     |
| DC     | direct current                      |
| EAR    | Equação Algébrica de Riccati        |
| GDL    | graus de liberdade                  |
| ISR    | interrupção de serviço e rotina     |
| LMI    | linear matrices inequalities        |
| MIMO   | multiple input multiple output      |
| PD     | proporcional derivativo             |
| PID    | proporcional integrativo derivativo |
| PTP    | point to point                      |
| UARMII | Underactuated Robot Manipulator II  |
| VSC    | variable structure control          |

# List of Symbols

|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| $a$              | Distância                                                              |
| $\mathbf{a}$     | Vetor de distâncias                                                    |
| $\mathbf{e}_j$   | Vetor unitário de dimensão $n$ e com o $j$ -ésimo componente igual a 1 |
| $\mathbf{K}$     | Matriz de rigidez                                                      |
| $m_1$            | Massa do cumpim                                                        |
| $\delta_{k-k_f}$ | Delta de Kronecker no instante $k_f$                                   |

# Contents

|       |                                                                         |    |
|-------|-------------------------------------------------------------------------|----|
| 1     | INTRODUCTION . . . . .                                                  | 15 |
| 1.1   | Motivation . . . . .                                                    | 15 |
| 1.2   | Contributions of the present work . . . . .                             | 16 |
| 1.3   | Outline of the remaining chapters . . . . .                             | 16 |
| 2     | DETERMINATION OF TARGET PERIODIC TRAJECTORIES . . . . .                 | 17 |
| 2.1   | Statement of the problem . . . . .                                      | 18 |
| 2.2   | Optimization . . . . .                                                  | 18 |
| 2.3   | Calculating $x_r$ . . . . .                                             | 18 |
| 2.4   | Application Examples . . . . .                                          | 20 |
| 2.4.1 | Variant Dynamic Circuit 1 . . . . .                                     | 20 |
| 2.4.2 | Variant Dynamic Circuit 2 . . . . .                                     | 20 |
| 3     | PREDICTIVE CONTROL FOR STABILIZATION OF PERIODIC TRAJECTORIES . . . . . | 21 |
| 3.1   | Statement of the problem . . . . .                                      | 21 |
| 3.2   | Linearization procedure . . . . .                                       | 23 |
| 3.3   | Optimal control problem . . . . .                                       | 32 |
| 3.3.1 | Cost function . . . . .                                                 | 32 |
| 3.3.2 | Prediction equation . . . . .                                           | 32 |
| 3.3.3 | Constraints . . . . .                                                   | 32 |
| 3.3.4 | Feasibility region . . . . .                                            | 32 |
| 3.4   | Application Examples . . . . .                                          | 32 |
| 3.4.1 | Integrador duplo . . . . .                                              | 32 |

---

|       |                                                  |    |
|-------|--------------------------------------------------|----|
| 3.4.2 | Patino 1 . . . . .                               | 32 |
| 3.4.3 | Patino 2 . . . . .                               | 32 |
| 4     | CONCLUSÃO . . . . .                              | 36 |
|       | BIBLIOGRAPHY . . . . .                           | 37 |
|       | APPENDIX A – TÓPICOS DE DILEMA LINEAR . . . . .  | 38 |
| A.1   | Modelo de simulação . . . . .                    | 38 |
| A.1.1 | get_x0 . . . . .                                 | 38 |
| A.1.2 | get_xf . . . . .                                 | 38 |
|       | ANNEX A – EXEMPLO DE UM PRIMEIRO ANEXO . . . . . | 39 |
| A.1   | Uma Seção do Primeiro Anexo . . . . .            | 39 |

# 1 Introduction

## 1.1 Motivation

The dynamics of switched affine systems can be described by a state equation of the form  $\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, t \in \mathbb{R}_+$ . Where  $x(t) \in \mathbb{R}^n$  is the state vector and  $\sigma(t) \in \{1, 2, \dots, N\}$  is the switching state, which is associated to the matrix  $A_{\sigma(t)} \in \mathbb{R}^{n \times n}$  and the vector  $b_{\sigma(t)} \in \mathbb{R}^n$  at each time  $t$ .

The use of switched affine systems to solve problems in engineering can be seen in fields like biochemical reactions, where Parise *et al.* (2018) uses switched affine systems to represent the moments of evolution in a generic biochemical reaction network; urban traffic, where Hajiahmadi *et al.* (2016) approaches the urban traffic control; power converters, where Benmiloud *et al.* (2019) also uses this approach and other areas where the behavior of the system can be described as a switched affine system.

There is extensive research in this class of controllers, and most of them solve the problem by driving the system states to the neighborhood of the target states, not considering the trajectory behavior of the system. This can reduce the performance of the system when the system is affected by the trajectory of the limit cycle in steady operation.

This problem has been the target of many studies, Benmiloud *et al.* (2019) uses a hybrid Poincaré map approach to stabilize a switched affine system into the desired limit cycle and uses as an example a DC-DC converter. by Egidio *et al.* (2020b), which proposes a control design of a state-dependent switching function for discrete-time switched affine systems to drive the system to the desired limit cycle that is defined by design associated with the expected performance of the system. The solution was applied in an academic numerical example and a theoretical DC-DC converter. In Egidio *et al.* (2020a) the author expanded the work to continuous-time switched affine systems.

Some studies form the basis for the present work:

Vieira (2013) solved the problem for a particular class of affine systems. Here the controller drove the system to a target state using MILP (Mixed-Integer Linear Programming) and the cyclic trajectory was a consequence of the states as the system approached

the target.

Patino *et al.* (2010) defined the target trajectory using nonlinear programming and used this target trajectory in the predictive control in terms of errors.

Marcolino (2018) proposed a solution with a controller law that drives the switched affine system with dwell-time constraints to a target cyclic trajectory. In this work, the problem is addressed in two stages: first, the problem is interpreted as a convex optimization where the author can find a targeted cyclic trajectory that minimizes a cost function associated with the system performance; then a second part that uses the target trajectory as input and computes perturbation at the switching instants of the reference control signal using predictive control.

## 1.2 Contributions of the present work

This present work proposes to expand the solution from )matheus2018 and provide a broader solution that can be used for systems that have their dynamics changing over time.

In a system that can be represented by a switched affine system with the form

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}$$

there are cases in which the dynamic response of the system can change depending on the operation mode. This can be represented in switched affine systems with proper matrices  $A_{\sigma(t)}$  and  $b_{\sigma(t)}$  for each operation mode. With the current implementation of Marcolino (2018) considering an invariant system during its operation, there is a need for expanding the solution so it can represent more systems and solve more engineering problems that can be expressed as switched affine systems.

## 1.3 Outline of the remaining chapters

The remaining of this text is organized as follows. Chapter 2 describes the problem of finding a target periodic trajectory for the given switched affine system. Chapter 3 proposes then a predictive control law that drives the system to the target cyclic trajectory. Finally, chapter 4 brings some remarks and considerations of the work.



## 2 Determination of Target Periodic Trajectories

Devemos encontrar uma trajetória cíclica que respeite todos as restrições do sistema e minimize a distancia em torno do valor de referencia?

Função custo:

$$J(\tau^\infty, \Omega^\infty, s^\infty) = \min_{\tau, \Omega, s} \int_0^{T_p} [x(t) - \bar{x}^\infty]^T Q [x(t) - \bar{x}^\infty] dt \quad (2.1)$$

As entradas para a otimização são:

- Lista de modos de operação
- Lista de matriz A associada ao modo de operação
- Lista de matriz B associada ao modo de operação
- Matriz C
- Matriz D
- Lista de valores de controle em cada modo
- Valor de referência dos estados (xref)
- Tempo máximo de trajetória cíclica (Tpmax)
- Número máximo de estados
- Peso dos estados na função custo (Q)

Em que  $Q = Q^T > 0$  e  $T_p$  é o período do ciclo.

A condição inicial é determinada de forma que o último ponto é coincidente com o primeiro ponto (dada a sequência de modos e instantes de chaveamento)

$$x(0) = (I - F_N F_{N-1} \dots F_1)^{-1} c \quad (2.2)$$

$$c = F_N F_{N-1} \dots F_2 G_1 u_1 + F_N F_{N-1} \dots F_3 G_2 u_2 + \dots + F_N G_{N-1} + G_N u_N \quad (2.3)$$

O resultado da otimizacao é  $\tau^\infty$  e  $x_0$

Exemplos:

Integrador (Matheus): Nao há otimizacao aqui, usamos a mesma trajetoria do Matheus para que tivessemos uma comparacao entre os resultados das simulacoes.

Patino, aqui estou usando um otimizador para encontrar a trajetoria de referencia.

Resultados:

Descrever resultados trajetorias Patino 1 e 2

**TODO: FAZER SIMULACAO COMPLETA DE CADA CASO**

## 2.1 Statement of the problem

## 2.2 Optimization

description of the optimization for finding the target trajectory

## 2.3 Calculating $x_r$

$$x_0 \in \mathfrak{R}$$

$$x(t) = e^{At} x(0) + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau \quad (2.4)$$

Modos:

$$\begin{aligned} [0 - t_1] : \dot{x} &= A_1 x + B_1 u_1 \\ [t_1 - t_2] : \dot{x} &= A_2 x + B_2 u_2 \\ &\vdots \\ [t_{N-1} - t_N] : \dot{x} &= A_N x + B_N u_N \end{aligned} \quad (2.5)$$

$$\begin{aligned}
x(t_1) &= e^{A_1 t_1} x(0) + \int_0^{t_1} e^{A_1(t_1-\tau)} B_1 u_1 d\tau \\
x(t_1) &= e^{A_1 t_1} x(0) + \left[ \int_0^{t_1} e^{A_1(t_1-\tau)} B_1 d\tau \right] u_1 \\
x(t_1) &= F_1 x(0) + G_1 u_1
\end{aligned} \tag{2.6}$$

$$x(t_1) = F_1 x(0) + G_1 u_1 \tag{2.7}$$

$$x(t_2) = F_2 x(1) + G_2 u_2 \tag{2.8}$$

$$\vdots$$

$$x(t_N) = F_N x(N-1) + G_N u_N \tag{2.9}$$

$$\begin{aligned}
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&\quad F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N) \\
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + c
\end{aligned} \tag{2.10}$$

Queremos  $x(t_N) = x(0) = ?$

$$\begin{aligned}
x(0) &= F_N F_{N-1} \dots F_1 x(0) + c \\
\underbrace{(I - F_N F_{N-1} \dots F_1)}_{\text{suposta invertível}} x(0) &= c \\
x(0) &= (I - F_N F_{N-1} \dots F_1)^{-1} c
\end{aligned} \tag{2.11}$$

em que

$$\begin{aligned}
c &= F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&\quad F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N)
\end{aligned} \tag{2.12}$$

## **2.4 Application Examples**

### **2.4.1 Variant Dynamic Circuit 1**

### **2.4.2 Variant Dynamic Circuit 2**

# 3 Predictive Control for Stabilization of Periodic Trajectories

## 3.1 Statement of the problem

considerando a planta de um sistema chaveado afim descrito por:

$$\dot{x}(t) = A_{\sigma}(t)x(t) + B_{\sigma}(t) \quad (3.1)$$

submetido a restrições de tempo de chaveamento.

A ação de controle aplicada no sistema é na forma de perturbacoes no tempo de chaveamento de cada modo de operação, e, para podermos controlar o sistema com controle preditivo, precisamos encontrar uma equacao linear para trabalhar a relação entre os instantes de chaveamento e o estado resultante do sistema ao final de um ciclo.

### PROBLEMA DE CONTROLE ÓTIMO

funcao custo

$$J = \sum_{i=1}^N \hat{e}_{k+i|k} Q \hat{e}_{k+i|k}^T + \rho \hat{\delta t}_{k+i|k}^2 \quad (3.2)$$

em que  $Q = Q^T > 0$ ,  $\rho > 0$

### EQUACAO DE PRECIÇÃO

$$\hat{x} = G \hat{\delta t} + f \quad (3.3)$$

$$\hat{\mathbf{x}} = \begin{bmatrix} \hat{\mathbf{x}}(t_{r_{k+1}} + \hat{\delta t}_{k+1|k}) \\ \hat{\mathbf{x}}(t_{r_{k+2}} + \hat{\delta t}_{k+1|k} + \hat{\delta t}_{k+2|k}) \\ \vdots \\ \hat{\mathbf{x}}(t_{r_{k+N}} + \hat{\delta t}_{k+1|k} + \dots + \hat{\delta t}_{k+N|k}) \end{bmatrix} \quad (3.4)$$

$$\hat{\delta t} = \begin{bmatrix} \hat{\delta t}_{k+1|k} \\ \hat{\delta t}_{k+2|k} \\ \vdots \\ \hat{\delta t}_{k+N|k} \end{bmatrix} \quad (3.5)$$

$$\mathbf{f} = \begin{bmatrix} \mathbf{f}_{k+1|k} \\ \mathbf{f}_{k+2|k} \\ \vdots \\ \mathbf{f}_{k+N|k} \end{bmatrix} \quad (3.6)$$

$$G = \begin{bmatrix} G_{1,1} & 0 & 0 & \dots & 0 \\ G_{2,1} & G_{2,2} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ G_{N,1} & G_{N,2} & G_{N,3} & \dots & G_{N,N} \end{bmatrix} \quad (3.7)$$

Now, let  $\mathbf{x}^r$  be the vector of target values defined as

$$\mathbf{x}^r = \begin{bmatrix} x_r(t_{r_{k+1}}) \\ x_r(t_{r_{k+2}}) \\ \vdots \\ x_r(t_{r_{k+N}}) \end{bmatrix} \quad (3.8)$$

In view of [completar texto aqui](#), the cost function is rewritten as

$$J = (\hat{\mathbf{x}} - \mathbf{x}^r)^T \mathbf{Q} (\hat{\mathbf{x}} - \mathbf{x}^r) + \rho \hat{\delta t}^T \hat{\delta t} \quad (3.9)$$

where  $\mathbf{Q} = \mathbf{Q}^T > 0$  is an  $nN \times nN$  block-diagonal matrix defined as  $\mathbf{Q} = \text{diag}(Q, Q, \dots, Q)$ .

### CONSTRAINTS

As restricoes do controlador devem descrever os limites de operacao da dinamica do modelo e ainda encontrar uma soluç o que leve o sistema para mais proximo da referencia.

### ESCREVER EQUACOES DE RESTRICAO

descrever que a trajetoria deve ser admissivel e forma com que o sistema calcula o

controle tambem deve ser admissivel

fazer um rapido comentario sobre a forma que essa restricao é trabalhada no capitulo 2

### FEASIBILITY REGION

nao aplicavel ????

como calcular ??? restricoes do problem de prog quad

sobre o controle

variacoes do instante de chaveadmento

mas envolvem o estado inicial do ciclo

sea dependencia das restricoes com respeito ao estao inicial for linear

entao podemos encarar o  $x_0$  como uma variavel junto com as variaveis de controle

com isso as restricoes descrevem um polietro aumenta do de controle e estado inicial

geometria computacional mpttoolbox

projeta a geometria no subespaco

pode haver melhoria com o ponto final livre

interessante para avaliar o desempenho do controlador na regio de factibilidade

### APPLICATION EXAMPLES

- Integrador duplo( matheus )
- Patino 1
- Patino 2

### CONCLUSION

mostrar o modelo em malha aberta indo para a sua trajetoria ciclica estavel junto com o modelo com controle ativo mostrando a convergencia mais rapida

## 3.2 Linearization procedure

The solution fo the state equation (??) at time  $t_N$ , starting from the initial condition  $x(t_0)$ , is of the form

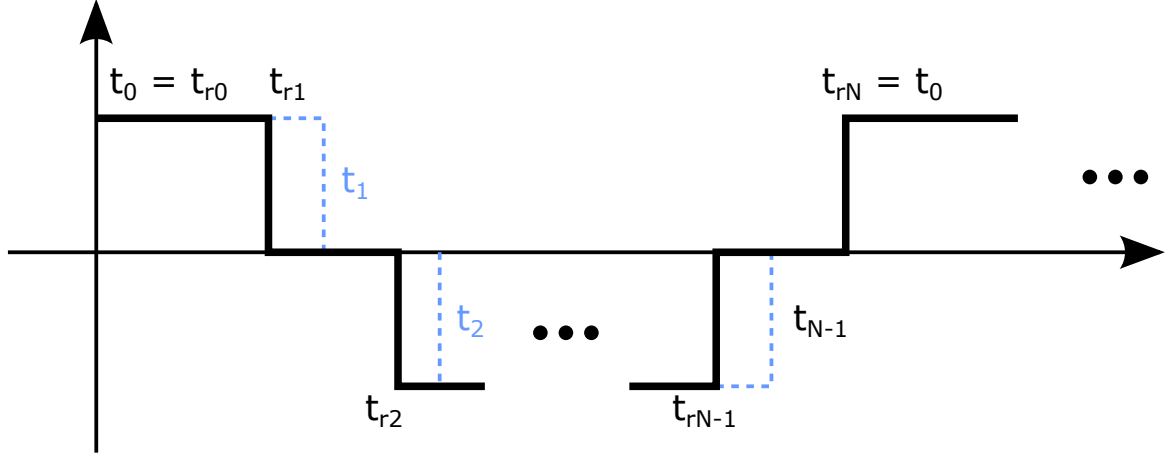


FIGURE 3.1 – Instantes de chaveamento

$$x(t_1) = e^{A_0(t_1-t_0)}x(t_0) + \int_{t_0}^{t_1} e^{A_0(t_1-\tau)} B_0 d\tau \quad (3.10)$$

$$x(t_{r_1}) = e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) + \int_{t_{r_0}}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \quad (3.11)$$

subtracting (3.10) and (3.11) we get

$$\begin{aligned} x(t_1) - x(t_{r_1}) &= \overbrace{e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0})}^{\square} \\ &+ \underbrace{\int_{t_0}^{t_1} e^{A_0(t_1-\tau)} B_0 d\tau - \int_{t_{r_0}}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau}_{\bigcirc} \end{aligned} \quad (3.12)$$

expanding  $\square$

$$\begin{aligned} \square &= e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) \\ &= e^{A_0(t_{r_1}+\delta t_1-t_0)}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) \\ &= e^{A_0(t_{r_1}-t_0)}e^{\delta t_1}x(t_0) - e^{A_0(t_{r_1}-t_{r_0})}x(t_{r_0}) \\ &= F_{r_0}e^{\delta t_1}x(t_0) - F_{r_0}x(t_{r_0}) \\ &= F_{r_0}(I + A_0\delta t_1)x(t_0) - F_{r_0}x(t_{r_0}) \\ &= F_{r_0}[x(t_0) - x_r(t_0)] + F_{r_0}A_0\delta t_1 \end{aligned} \quad (3.13)$$

expanding  $\bigcirc$



$$\begin{aligned}
 \bigcirc &= \int_{t_0}^{t_1} e^{A_0(t_1-\tau)} B_0 d\tau - \int_{t_{r_0}}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \\
 &= \int_{t_0}^{t_{r_1}+\delta t_1} e^{A_0(t_{r_1}+\delta t_1-\tau)} B_0 d\tau - \int_{t_0}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \\
 &= e^{A_0\delta t_1} \left[ \int_{t_0}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau + \int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \right] - \int_{t_0}^{t_{r_1}} e^{A_0(t_{r_1}-\tau)} B_0 d\tau \\
 &= e^{A_0\delta t_1} [G_{r_0} + B_0\delta t_1] - G_{r_0} \\
 &= (I + A_0\delta t_1)[G_{r_0} + B_0\delta t_1] - G_{r_0} \\
 &= \cancel{G_{r_0}} + B_0\delta t_1 + A_0G_{r_0}\delta t_1 + \cancel{A_0B_0\delta t_1\delta t_1} - \cancel{G_{r_0}} \\
 &= A_0G_{r_0}\delta t_1 + B_0\delta t_1
 \end{aligned} \tag{3.14}$$

and replacing (3.13) and (3.14) into equation (3.12) we get to a equation that computes  $x(t_1) - x_r(t_{r_1})$  from  $x(t_0) - x_r(t_{r_0})$

$$x(t_1) - x_r(t_{r_1}) = F_{r_0}[x(t_0) - x_r(t_0)] + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0]\delta t_1 \tag{3.15}$$

Repeating the procedare for  $x(t_2)$  we have

$$x(t_2) = e^{A_1(t_2-t_1)}x(t_1) + \int_{t_1}^{t_2} e^{A_1(t_2-\tau)} B_1 d\tau \tag{3.16}$$

$$x(t_{r_2}) = e^{A_1(t_{r_2}-t_{r_1})}x(t_{r_1}) + \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)} B_1 d\tau \tag{3.17}$$

subtracting 3.16 and 3.17

$$\begin{aligned}
 x(t_2) - x(t_{r_2}) &= \overbrace{e^{A_1(t_2-t_1)}x(t_1) - e^{A_1(t_{r_2}-t_{r_1})}x(t_{r_1})}^{\square} \\
 &\quad + \underbrace{\int_{t_1}^{t_2} e^{A_1(t_2-\tau)} B_1 d\tau - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)} B_1 d\tau}_{\bigcirc}
 \end{aligned} \tag{3.18}$$

expanding  $\square$

$$\begin{aligned}
 \square &= e^{A_1(t_{r_2}+\delta t_2-t_{r_1}-\delta t_1)}x(t_1) - e^{A_1(t_{r_2}-t_{r_1})}x_r(t_{r_1}) \\
 &= e^{A_1(t_{r_2}-t_{r_1})}e^{A_1(\delta t_2-\delta t_1)}x(t_1) - e^{A_1(t_{r_2}-t_{r_1})}x_r(t_{r_1}) \\
 &= F_{r_1}[I + A_1(\delta t_2 - \delta t_1)]x(t_1) - F_{r_1}x_r(t_{r_1}) \\
 &= F_{r_1}[x(t_1) - x_r(t_{r_1})] + F_{r_1}A_1x(t_1)\delta t_2 - F_{r_1}A_1x(t_1)\delta t_1
 \end{aligned} \tag{3.19}$$

expanding  $\bigcirc$

$$\begin{aligned}
 \bigcirc &= \int_{t_1}^{t_2} e^{A_1(t_2-\tau)}B_1d\tau - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau \\
 &= \int_{t_{r_1}+\delta t_1}^{t_{r_2}+\delta t_2} e^{A_1(t_{r_2}+\delta t_2-\tau)}B_1d\tau - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau \\
 &= e^{A_1\delta t_2} \left[ \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau - \int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_1(t_{r_2}-\tau)}B_1d\tau + \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_1(t_{r_2}-\tau)}B_1d\tau \right] \\
 &\quad - \int_{t_{r_1}}^{t_{r_2}} e^{A_1(t_{r_2}-\tau)}B_1d\tau \\
 &= (I + A_1\delta t_2) \left[ G_{r_1} - \int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_1(t_{r_2}-\tau)}B_1d\tau + \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_1(t_{r_2}-\tau)}B_1d\tau \right] - G_{r_1} \\
 &= (I + A_1\delta t_2) \left[ G_{r_1} - e^{A_1(t_{r_2}-t_{r_1})}B_1\delta t_1 + B_1\delta t_2 \right] - G_{r_1} \\
 &= (I + A_1\delta t_2) \left[ G_{r_1} - F_{r_1}B_1\delta t_1 + B_1\delta t_2 \right] - G_{r_1} \\
 &= \cancel{G_{r_1}} - F_{r_1}B_1\delta t_1 + B_1\delta t_2 + A_1G_{r_1}\delta t_2 - \cancel{G_{r_1}} \\
 &= A_1G_{r_1}\delta t_2 + B_1\delta t_2 - F_{r_1}B_1\delta t_1
 \end{aligned} \tag{3.20}$$

replacing (3.19) and (3.20) into equation (3.18)

$$\begin{aligned}
 x(t_2) - x_r(t_{r_2}) &= F_{r_1}[x(t_1) - x_{r_1}(t_{r_1})] \\
 &\quad - F_{r_1}[A_1x(t_1) + B_1]\delta t_1 \\
 &\quad + [F_{r_1}A_1 \underbrace{x(t_1)}_{\triangle} + A_1G_{r_1} + B_1]\delta t_2
 \end{aligned} \tag{3.21}$$

expanding  $\triangle$  from 3.15

$$\triangle = x(t_1) = x_r(t_{r_1}) + F_{r_0}[x(t_0) - x_r(t_0)] + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0]\delta t_1$$

note that when  $\triangle$  is replaced in eq. 3.21 it will produce a product with  $\delta t_1$  and  $\delta t_2$  like

$$x(t_1)\delta t_1 = x_r(t_{r_1})\delta t_1 + F_{r_0} \overbrace{[x(t_0) - x_r(t_0)]\delta t_1}^{\square} + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0] \overbrace{\delta t_1\delta t_1}^{\circ}$$

both  $\square$  and  $\circ$  are products of differences that produces a smaller number and they will computed as  $\approx 0$

$$x(t_1)\delta t_1 \approx x_r(t_{r_1})\delta t_1 \quad (3.22)$$

replacing equation 3.15 and 3.22 in 3.21

$$\begin{aligned} x(t_2) - x_r(t_{r_2}) &= F_{r_1} \{ F_{r_0} [x(t_0) - x_r(t_0)] + [F_{r_0}A_0x(t_0) + A_0G_{r_0} + B_0]\delta t_1 \} \\ &\quad - F_{r_1} [A_1x_r(t_1) + B_1]\delta t_1 \\ &\quad + [F_{r_1}A_1x_r(t_1) + A_1G_{r_1} + B_1]\delta t_2 \\ x(t_2) - x_r(t_{r_2}) &= F_{r_1}F_{r_0}[x(t_0) - x_r(t_0)] \\ &\quad + F_{r_1}[F_{r_0}A_0x(t_0) + A_0G_{r_0} - A_1x_r(t_{r_1}) + B_0 - B_1]\delta t_1 \\ &\quad + [F_{r_1}A_1x_r(t_{r_1}) + A_1G_{r_1} + B_1]\delta t_2 \end{aligned} \quad (3.23)$$

Now, repeating the procedure for  $x(t_3)$  we have

$$x(t_3) = e^{A_2(t_3-t_2)}x(t_2) + \int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2d\tau \quad (3.24)$$

$$x(t_{r_3}) = e^{A_2(t_{r_3}-t_{r_2})}x(t_{r_2}) + \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau \quad (3.25)$$

subtracting 3.24 and 3.25

$$\begin{aligned} x(t_3) - x(t_{r_3}) &= \overbrace{e^{A_2(t_3-t_2)}x(t_2) - e^{A_2(t_{r_3}-t_{r_2})}x(t_{r_2})}^{\square} \\ &\quad + \underbrace{\int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2d\tau - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau}_{\circ} \end{aligned} \quad (3.26)$$

expanding  $\square$

$$\begin{aligned}
 \square &= e^{A_2(t_{r_3}+\delta t_3-t_{r_2}-\delta t_2)}x(t_2) - e^{A_2(t_{r_3}-t_{r_2})}x_r(t_{r_2}) \\
 &= e^{A_2(t_{r_3}-t_{r_2})}e^{A_2(\delta t_3-\delta t_2)}x(t_2) - e^{A_2(t_{r_3}-t_{r_2})}x_r(t_{r_2}) \\
 &= F_{r_2}[I + A_2(\delta t_3 - \delta t_2)]x(t_2) - F_{r_2}x_r(t_{r_2}) \\
 &= F_{r_2}[x(t_2) - x_r(t_{r_2})] + F_{r_2}A_2x(t_2)\delta t_3 - F_{r_2}A_2x(t_2)\delta t_2
 \end{aligned} \tag{3.27}$$

expanding  $\bigcirc$

$$\begin{aligned}
 \bigcirc &= \int_{t_2}^{t_3} e^{A_2(t_3-\tau)}B_2d\tau - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau \\
 &= \int_{t_{r_2}+\delta t_2}^{t_{r_3}+\delta t_3} e^{A_2(t_{r_3}+\delta t_3-\tau)}B_2d\tau - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau \\
 &= e^{A_2\delta t_3} \left[ \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau - \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_2(t_{r_3}-\tau)}B_2d\tau + \int_{t_{r_3}}^{t_{r_3}+\delta t_3} e^{A_2(t_{r_3}-\tau)}B_2d\tau \right] \\
 &\quad - \int_{t_{r_2}}^{t_{r_3}} e^{A_2(t_{r_3}-\tau)}B_2d\tau \\
 &= (I + A_2\delta t_3) \left[ G_{r_2} - \int_{t_{r_2}}^{t_{r_2}+\delta t_2} e^{A_2(t_{r_3}-\tau)}B_2d\tau + \int_{t_{r_3}}^{t_{r_3}+\delta t_3} e^{A_2(t_{r_3}-\tau)}B_2d\tau \right] - G_{r_2} \\
 &= (I + A_2\delta t_3) \left[ G_{r_2} - e^{A_2(t_{r_3}-t_{r_2})}B_2\delta t_2 + B_2\delta t_3 \right] - G_{r_2} \\
 &= (I + A_2\delta t_3) \left[ G_{r_2} - F_{r_2}B_2\delta t_2 + B_2\delta t_3 \right] - G_{r_2} \\
 &= \cancel{G_{r_2}} - F_{r_2}B_2\delta t_2 + B_2\delta t_3 + A_2G_{r_2}\delta t_3 - \cancel{G_{r_2}} \\
 &= A_2G_{r_2}\delta t_3 + B_2\delta t_3 - F_{r_2}B_2\delta t_2
 \end{aligned} \tag{3.28}$$

replacing (3.27) in (3.28) back in eq. (3.26)

$$\begin{aligned}
 x(t_3) - x_r(t_{r_3}) &= F_{r_2}[x(t_2) - x_{r_2}(t_{r_2})] \\
 &\quad - F_{r_2}[A_2x(t_2) + B_2]\delta t_2 \\
 &\quad + [F_{r_2}A_2x(t_2) + A_2G_{r_2} + B_2]\delta t_3
 \end{aligned}$$

$$\begin{aligned}
 x(t_3) - x_r(t_{r_3}) = & F_{r_2} \{ F_{r_1} F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_1} [F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + [F_{r_1} A_1 x_r(t_1) + A_1 G_{r_1} + B_1] \delta t_2 \} \\
 & - F_{r_2} [A_2 x_r(t_2) + B_2] \delta t_2 \\
 & + [F_{r_2} A_2 x_r(t_2) + A_2 G_{r_2} + B_2] \delta t_3
 \end{aligned}$$

$$\begin{aligned}
 x(t_3) - x_r(t_{r_3}) = & F_{r_2} F_{r_1} F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_2} F_{r_1} [F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + F_{r_2} [F_{r_1} A_1 x_r(t_1) + A_1 G_{r_1} - A_2 x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & + [F_{r_2} A_2 x_r(t_2) + A_2 G_{r_2} + B_2] \delta t_3
 \end{aligned} \tag{3.29}$$

Continuing on expanding the propagation of the system further more we can deduce the equation for  $x(t_i)$  as

$$\begin{aligned}
 x(t_i) - x_r(t_{r_i}) = & F_{r_{i-1}} F_{r_{i-2}} \dots F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_{i-1}} F_{r_{i-2}} \dots F_{r_1} [F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + F_{r_{i-1}} F_{r_{i-2}} \dots F_{r_2} [F_{r_1} A_1 x_r(t_1) + A_1 G_{r_1} - A_2 x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + F_{r_{i-1}} [F_{r_{i-2}} A_{i-2} x_r(t_{i-2}) + A_{i-2} G_{r_{i-2}} - A_{i-1} x_r(t_{r_{i-1}}) + B_{i-2} - B_{i-1}] \delta t_{i-1} \\
 & + [F_{r_{i-1}} A_{i-1} x_r(t_{i-1}) + A_{i-1} G_{r_{i-1}} + B_{i-1}] \delta t_i
 \end{aligned} \tag{3.30}$$

and then, for  $x(t_N)$  we have

$$\begin{aligned}
 x(t_N) - x_r(t_{r_N}) = & F_{r_{N-1}} F_{r_{N-2}} \cdots F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_{N-1}} F_{r_{N-2}} \cdots F_{r_1} \overbrace{[F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1]}^{\square} \delta t_1 \\
 & + F_{r_{N-1}} F_{r_{N-2}} \cdots F_{r_2} [F_{r_1} A_1 x(t_1) + A_1 G_{r_1} - A_2 x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + F_{r_{N-1}} [F_{r_{N-2}} A_{N-2} x_r(t_{N-2}) + A_{N-2} G_{r_{N-2}} - A_{N-1} x_r(t_{r_{N-1}}) \\
 & \quad + B_{N-2} - B_{N-1}] \delta t_{N-1} \\
 & + \underbrace{[F_{r_{N-1}} A_{N-1} x_r(t_{N-1}) + A_{N-1} G_{r_{N-1}} + B_{N-1}]}_{\bigcirc} \delta t_N
 \end{aligned} \tag{3.31}$$

$$\begin{aligned}
 \square = & F_{r_0} A_0 x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & A_0 F_{r_0} x(t_0) + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & A_0 [x_r(t_{r_1}) - G_{r_0}] + A_0 G_{r_0} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & A_0 x_r(t_{r_1}) - \cancel{A_0 G_{r_0}} + \cancel{A_0 G_{r_0}} - A_1 x_r(t_{r_1}) + B_0 - B_1 \\
 = & (A_0 - A_1) x_r(t_{r_1}) + B_0 - B_1
 \end{aligned} \tag{3.32}$$

$$\begin{aligned}
 \bigcirc = & F_{r_{N-1}} A_{N-1} x_r(t_{N-1}) + A_{N-1} G_{r_{N-1}} + B_{N-1} \\
 = & A_{N-1} F_{r_{N-1}} x_r(t_{N-1}) + A_{N-1} G_{r_{N-1}} + B_{N-1} \\
 = & A_{N-1} [x_r(t_{r_N}) - G_{r_{N-1}}] + A_{N-1} G_{r_{N-1}} + B_{N-1} \\
 = & A_{N-1} x_r(t_{r_N}) - \cancel{A_{N-1} G_{r_{N-1}}} + \cancel{A_{N-1} G_{r_{N-1}}} + B_{N-1} \\
 = & A_{N-1} x_r(t_{r_N}) + B_{N-1}
 \end{aligned} \tag{3.33}$$

replacing 3.32 and 3.33 in 3.31

$$\begin{aligned}
 x(t_N) - x_r(t_{r_N}) = & F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_0} [x(t_0) - x_r(t_0)] \\
 & + F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_1} [(A_0 - A_1)x_r(t_{r_1}) + B_0 - B_1] \delta t_1 \\
 & + F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_2} [(A_1 - A_2)x_r(t_{r_2}) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + F_{r_{N-1}} [(A_{N-2} - A_{N-1})x_r(t_{r_{N-1}}) + B_{N-2} - B_{N-1}] \delta t_{N-1} \\
 & + [A_{N-1}x_r(t_{r_N}) + B_{N-1}] \delta t_N
 \end{aligned} \tag{3.34}$$

$$e(t) = x(t) - x_r(t_r) \tag{3.35}$$

$$\Phi = F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_0} \tag{3.36}$$

$$\begin{aligned}
 \Gamma = & \begin{bmatrix} F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_1} [(A_0 - A_1)x_r(t_{r_1}) + B_0 - B_1], \\ F_{r_{N-1}} F_{r_{N-2}} \dots F_{r_2} [(A_1 - A_2)x_r(t_{r_2}) + B_1 - B_2], \\ \vdots \\ F_{r_{N-1}} [(A_{N-2} - A_{N-1})x_r(t_{r_{N-1}}) + B_{N-2} - B_{N-1}] \end{bmatrix}
 \end{aligned} \tag{3.37}$$

$$\delta t = [\delta t_1, \delta t_2, \dots, \delta t_N]^T \tag{3.38}$$

$$e(t_N) = \Phi e(t_0) + \Gamma \delta t \tag{3.39}$$

where  $x_r$  is calculated as

$$x_r(t_{r_1}) = F_0 x_r(t_{r_0}) + G_0 \tag{3.40}$$

$$x_r(t_{r_2}) = F_1 x_r(t_{r_1}) + G_1 \tag{3.41}$$

$$\vdots$$

$$x_r(t_{r_N}) = F_{N-1} x_r(t_{r_{N-1}}) + G_{N-1} \tag{3.42}$$

and  $F_i$  and  $G_i$  are calculated as

$$[F_i, G_i] = c2dm(Ai, Bi, [], [], dti, 'zoh') \tag{3.43}$$

## 3.3 Optimal control problem

### 3.3.1 Cost function

### 3.3.2 Prediction equation

### 3.3.3 Constraints

### 3.3.4 Feasibility region

## 3.4 Application Examples

### 3.4.1 Integrador duplo

### 3.4.2 Patino 1

### 3.4.3 Patino 2

Matrizes do artigo patino exemplo 2:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & -\frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}x_3(t) \end{bmatrix} \quad (3.44)$$

$$A = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & \frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \quad (3.45)$$



$$\begin{aligned}
 \mathbf{u}_0 &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T \\
 \mathbf{u}_1 &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T \\
 \mathbf{u}_2 &= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T \\
 \mathbf{u}_3 &= \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^T \\
 \mathbf{u}_4 &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T \\
 \mathbf{u}_5 &= \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T \\
 \mathbf{u}_6 &= \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^T \\
 \mathbf{u}_7 &= \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T
 \end{aligned} \tag{3.46}$$

$$A_{u_0} = A\mathbf{u}_0 = \begin{bmatrix} 0 \\ 0 \\ \frac{-Rx_3(t)}{L} \end{bmatrix} \tag{3.47}$$

$$A_{u_1} = A\mathbf{u}_1 = \begin{bmatrix} 0 \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.48}$$

$$A_{u_2} = A\mathbf{u}_2 = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ \frac{-x_3(t)}{C_2} \\ \frac{x_2(t)-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.49}$$

$$A_{u_3} = A\mathbf{u}_3 = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ 0 \\ \frac{E-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.50}$$

$$A_{u_4} = A\mathbf{u}_4 = \begin{bmatrix} \frac{-x_3(t)}{C_1} \\ 0 \\ \frac{x_1(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.51}$$

$$A_{u_5} = A\mathbf{u}_5 = \begin{bmatrix} \frac{-x_3(t)}{C_1} \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_1(t)-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \tag{3.52}$$

$$A_{u_6} = A\mathbf{u}_6 = \begin{bmatrix} 0 \\ \frac{-x_3(t)}{C_2} \\ \frac{x_2(t) - Rx_3(t)}{L} \end{bmatrix} \quad (3.53)$$

$$A_{u_7} = A\mathbf{u}_7 = \begin{bmatrix} 0 \\ 0 \\ \frac{E - Rx_3(t)}{L} \end{bmatrix} \quad (3.54)$$

$$\dot{X} = A_{u_0}X + B_0u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.55)$$

$$\dot{X} = A_{u_1}X + B_1u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.56)$$

$$\dot{X} = A_{u_2}X + B_2u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & -\frac{1}{C_2} \\ -\frac{1}{L} & \frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.57)$$

$$\dot{X} = A_{u_3}X + B_3u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.58)$$

$$\dot{X} = A_{u_4}X + B_4u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & 0 \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.59)$$

$$\dot{X} = A_{u_5}X + B_5u = \begin{bmatrix} 0 & 0 & -\frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ \frac{1}{L} & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.60)$$

$$\dot{X} = A_{u_6}X + B_6u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (3.61)$$

$$\dot{X} = A_{u_7}X + B_7u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (3.62)$$



# 4 Conclusão

Concluimos que esse eh o capitulo 4...

colocar um plano de trabalho e cronograma

possibilidade de uma aplicacao do trabalho usando um conversor de tensao

|                                        | 2022 |   |   |   | 2023 |   |  |  |
|----------------------------------------|------|---|---|---|------|---|--|--|
|                                        | 1    | 2 | 1 | 2 | 1    | 2 |  |  |
| extensão de caso tempo final           |      |   |   |   |      |   |  |  |
| elaboração de um artigo com resultados |      |   |   |   |      |   |  |  |
| aplicação com maior número de estados  |      |   |   |   |      |   |  |  |
| experimental                           |      |   |   |   |      |   |  |  |
| elaboração de tese final(defesa)       |      |   |   |   |      |   |  |  |

# Bibliography

BENMILOUD, M.; BENALIA, A.; DJEMAI, M.; DEFOORT, M. On the local stabilization of hybrid limit cycles in switched affine systems. **IEEE Transactions on Automatic Control**, v. 64, n. 2, p. 841–846, 2019.

EGIDIO, L.; DEAECTO, G.; GEROMEL, J. Limit cycle global asymptotic stability of continuous-time switched affine systems. **IFAC-PapersOnLine**, v. 53, p. 6121–6126, 01 2020.

EGIDIO, L. N.; DAIHA, H. R.; DEAECTO, G. S. Global asymptotic stability of limit cycle and  $h_2/h_\infty$  performance of discrete-time switched affine systems. **Automatica**, v. 116, p. 108927, 2020. ISSN 0005-1098.

HAJIAHMADI, M.; SCHUTTER, B. D.; HELLENDORRN, H. Robust  $h_\infty$  switching control techniques for switched nonlinear systems with application to urban traffic control. **International Journal of Robust and Nonlinear Control**, Wiley Online Library, v. 26, n. 6, p. 1286–1306, 2016.

MARCOLINO, M. H. **Stabilization of Periodic trajectories through receding horizon optimization of switching times**. 2018. 79 f. Tese (Doutorado em Sistemas de Controle) — Instituto Tecnológico de Aeronáutica, São José dos Campos, 2018.

PARISE, F.; VALCHER, M. E.; LYGEROS, J. Computing the projected reachable set of stochastic biochemical reaction networks modeled by switched affine systems. **IEEE Transactions on Automatic Control**, v. 63, n. 11, p. 3719–3734, 2018.

PATINO, D.; RIEDINGER, P.; RUIZ, F. A predictive control approach for dc-dc power converters and cyclic switched systems. In: IEEE. **2010 IEEE International Conference on Industrial Technology**. [S.l.], 2010. p. 1259–1264.

VIEIRA, M. S. **Controle Preditivo de sistemas com atuadores sujeitos a restrições temporais de chaveamento**. 2013. 119 f. Dissertação (Mestrado em Sistemas de Controle) — Instituto Tecnológico de Aeronáutica, São José dos Campos, 2013.

# Appendix A - Tópicos de Dilema Linear

## A.1 Modelo de simulação

```
+engine
├─ CORES.m
├─ fun_custo_patino.m
├─ get_config_sim_matheus.m
├─ get_config_sim_patino_1.m
├─ get_config_sim_patino_2.m
├─ get_config.m
├─ get_otmin_opt.m
├─ get_ts.m
├─ get_x0.m
├─ get_xr.m
├─ nonlincon_patino.m
├─ otmin.m
├─ sim_1.m
├─ sim_n.m
└─ +mpc
    ├─ construcao_modelo_instantes.m
    ├─ matrizes_ss_mpc_dualmode_switching.m
    └─ mpc_dualmode_switching.m
```

### A.1.1 get\_x0

funcao de calculo da condicao inicial

### A.1.2 get\_xf

Texto do modelo de simulação sendo colocado aqui.

# **Annex A - Exemplo de um Primeiro Anexo**

## **A.1 Uma Seção do Primeiro Anexo**

Algum texto na primeira seção do primeiro anexo.

## FOLHA DE REGISTRO DO DOCUMENTO

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                |                                         |                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------|------------------------|
| 1. CLASSIFICAÇÃO/TIPO<br>TD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2. DATA<br>25 de março de 2015 | 3. DOCUMENTO Nº<br>DCTA/ITA/DM-018/2015 | 4. Nº DE PÁGINAS<br>39 |
| 5. TÍTULO E SUBTÍTULO:<br>Robust control of linear systems with Switched Actuators Subjected to Dwell-Time Constraints                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                |                                         |                        |
| 6. AUTOR(ES):<br><b>Daniel Vieira</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                |                                         |                        |
| 7. INSTITUIÇÃO(ÕES)/ÓRGÃO(S) INTERNO(S)/DIVISÃO(ÕES):<br>Instituto Tecnológico de Aeronáutica – ITA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |                                         |                        |
| 8. PALAVRAS-CHAVE SUGERIDAS PELO AUTOR:<br>Cupim; Cimento; Estruturas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                |                                         |                        |
| 9. PALAVRAS-CHAVE RESULTANTES DE INDEXAÇÃO:<br>Cupim; Dilema; Construção                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |                                         |                        |
| 10. APRESENTAÇÃO: (X) Nacional ( ) Internacional<br>ITA, São José dos Campos. Curso de Mestrado. Programa de Pós-Graduação em Engenharia Aeronáutica e Mecânica. Área de Sistemas Aeroespaciais e Mecatrônica. Orientador: Prof. Dr. Adalberto Santos Dupont. Coorientadora: Prof <sup>a</sup> . Dr <sup>a</sup> . Doralice Serra. Defesa em 05/03/2015. Publicada em 25/03/2015.                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                |                                         |                        |
| 11. RESUMO:<br>Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o systema à trajetória cíclica alvo. |                                |                                         |                        |
| 12. GRAU DE SIGILO:<br>(X) OSTENSIVO ( ) RESERVADO ( ) SECRETO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                |                                         |                        |